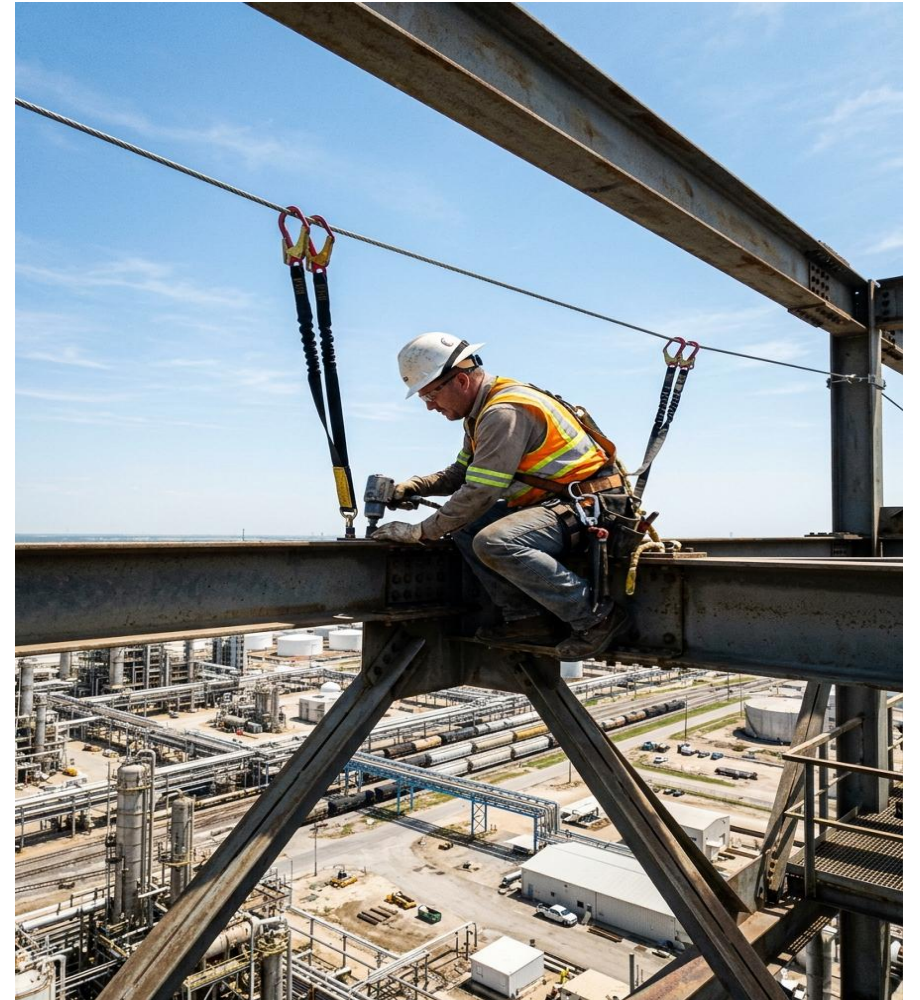


TRAINING

Work at Height

Practical basics of safe work at height, choosing fall arrest systems, rescue and first aid

Training course for workers who perform work at height and for line supervisors



After the course you can

- 01** — Recognise the hazards of work at height, assess risks and meet the conditions for admission to work.
- 02** — Correctly choose and use restraint and fall arrest systems, anchor devices and ladders.
- 03** — Check the harness before work and follow the prohibitions for work at height.
- 04** — Act instantly in incidents: evacuate a suspended worker and give first aid.

**The course's main goal
— zero injuries and
prevention of falls from
height.**

Four course topics

- 01 — Criteria for work at height and admission to it.
- 02 — Fall protection systems, anchors, ladders and harnesses.
- 03 — Risks and prohibitions in work at height.
- 04 — Combined work, suspension rescue and first aid.

COURSE

4

topics of safe work at
height



Work at height and steepjack work

Falls from height are the leading cause of fatal workplace injuries. The right choice of fall protection saves lives.

Reference: Rules on occupational safety for work at height

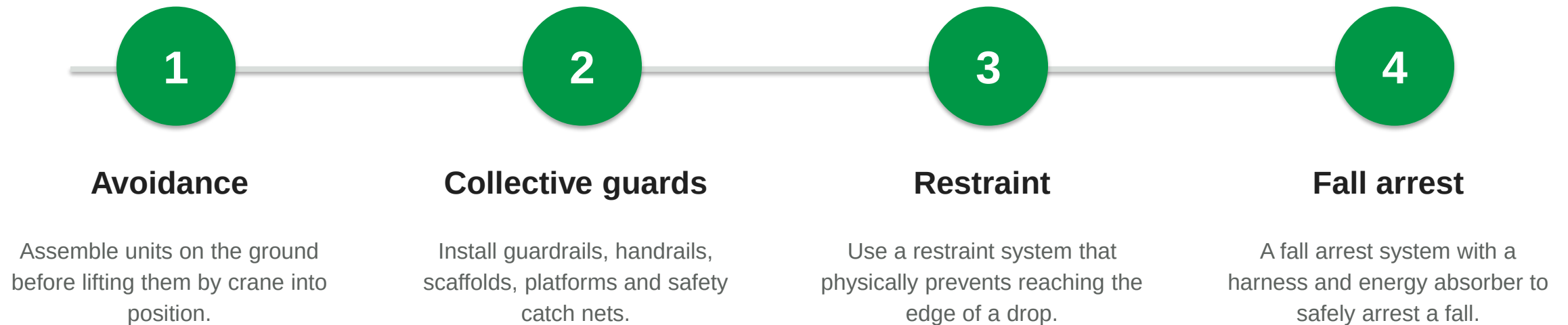


Work at height defined: regulatory criteria

- **Basic height criterion:** work at a height of 1.3 m or more above the ground, a floor slab or a work deck.
- **Guarding:** work on platforms within 2 m of unguarded drops of 1.3 m or more, or where guardrails are lower than 1.1 m.
- **Hazards below:** work less than 1.3 m above machines, mechanisms, protruding objects, water or molten metal.
- **High-level (steeplejack) work:** work more than 5 m above the ground or a floor slab, done from erection fixtures or directly from structures.

Even an unprotected fall from a height of 1.3 metres can end in severe injury or death!

Fall protection hierarchy: 4 levels



Fall protection PPE is the last line of defence! Collective guards always come first!

Admission to work at height

- **Age and health:** only persons aged 18 or over who have passed a mandatory medical examination are admitted to work at height.
- **Training:** the worker is trained in occupational safety and health and holds a knowledge test certificate; knowledge is tested at least once a year.
- **First admission:** a worker admitted to work at height for the first time receives instruction from the responsible work manager over 5 work shifts.
- **Crew and permit:** work at height is done under a work permit by a crew (team) of at least 2 people.

Working at height alone is prohibited — only in a crew of at least 2 people and under a work permit!

Two protection systems: Restraint vs Fall arrest



RESTRAINT SYSTEM

- Goal: physically PREVENT access to the edge of a drop.
- Components: a work belt or harness + a fixed-length lanyard.
- No fall: the fall factor is zero (no dynamic shock load).
- Load: no energy absorber needed, as a fall is impossible.
- Limitation: never use where a fall into a void is possible!



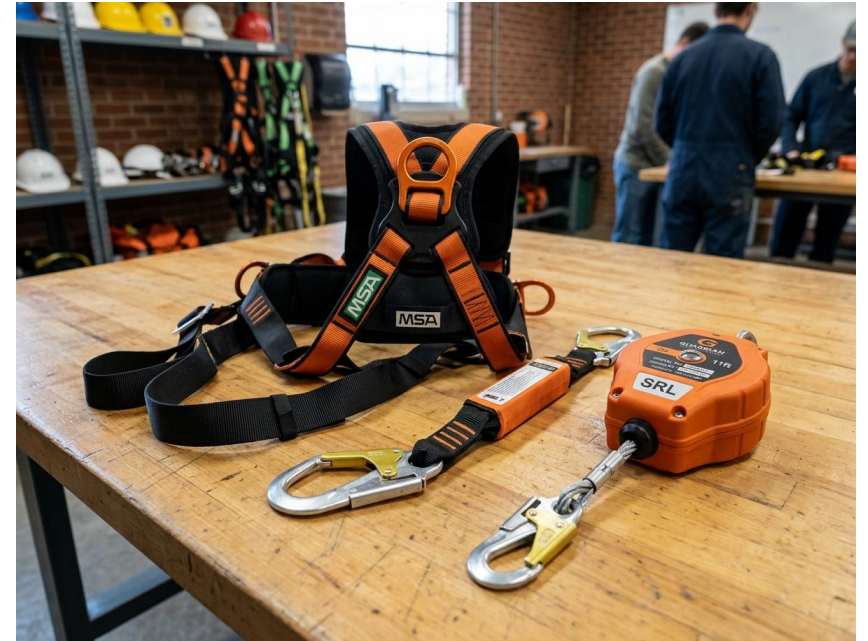
FALL ARREST SYSTEM

- Goal: SAFELY ARREST a fall that has already begun.
- Components: full body harness + energy absorber + anchor line.
- Shock absorption: the absorber cuts the arrest force to 6 kN or less.
- Load distribution: the load is transferred to the thighs, pelvis and chest.
- Prerequisite: calculate the clearance below the worker.

A work positioning belt (without shoulder and thigh straps) is PROHIBITED as a fall arrest system!

Fall arrest system components: The ABC rule

- **A (Anchor):** a reliable anchor device (sling, clamp, beam clamp) with a strength of at least 22 kN (2.2 tonnes).
- **B (Body wear):** a full body harness with shoulder and thigh straps and a dorsal fall arrest D-ring.
- **C (Connector):** a lanyard with an energy absorber, a self-retracting lifeline (SRL) or a rope grab on a flexible line.
- **100% tie-off rule:** when moving along structures, use a twin-leg (Y-shaped) lanyard — one leg is always clipped in!



Fall protection PPE works only with correct anchorage and a calculated clearance!

Fall factor and clearance

- Fall factor = Fall height / Lanyard length. It determines how hard the dynamic shock on the worker's body is in a fall.
- **Factor 0 (anchor above head):** the best option, minimal fall distance, gentle absorber deployment.
- **Factor 1 (anchor at chest/shoulder level):** a standard fall of the slack lanyard length; acceptable with an energy absorber.
- **Factor 2 (anchor at foot level):** CRITICAL risk! Free fall equals twice the lanyard length (up to 4 m); risk of hitting structures.
- **Clearance:** the calculated distance below the workplace (lanyard length + absorber extension 1.2 m + height 1.8 m + margin 1.0 m = 5.7–6.0 m).

**Clip the lanyard AS HIGH
ABOVE YOUR HEAD AS
POSSIBLE!**

**Factor 2 — extreme load
on the anchor and spine!**

Anchor devices and rigid / flexible lines

- **Class A:** fixed structural anchor points on concrete and steel load-bearing building structures.
- **Class B:** portable temporary anchor devices (textile slings, tripods, I-beam clamps).
- **Class C:** flexible horizontal cable anchor lines for free, continuous movement of the worker along the structure.
- **Class D:** rigid horizontal rail anchor lines with a sliding trolley for heavy-duty use.
- **Never anchor to unreliable supports:** process pipelines, cable trays, radiators and window frames!

Never attach the lanyard to process pipes or cable trays — only to verified load-bearing structures!

Ladders, stepladders and access platforms

- **Three points of contact rule:** when climbing up or down a ladder, always keep two hands and one foot (or two feet and one hand) on it.
- **Leaning ladder angle:** strictly 70–75 degrees to the horizontal (height-to-base ratio 4:1).
- **Top end:** a leaning ladder must extend at least 1.0 metre above the landing.
- **Work ban:** do not work from the top two steps of stepladders without handrails and guarded platforms.
- **No heavy work:** heavy erection and power electrical installation work must not be done from leaning ladders.

A leaning ladder is a means of access, not a workplace!
Heavy work is done only from scaffolds and platforms!



4 steps of a pre-use harness check

1

Webbing

Check the webbing for cuts, burns, abrasion and intact contrast stitching.

2

Metal hardware

Inspect D-rings and buckles for deformation, cracks, corrosion and secure latches.

3

Fall indicator

Make sure the fabric fall indicator is intact (the harness has not arrested a fall).

4

Body fit

Adjust the thigh and shoulder straps: a flat hand should fit snugly under the thigh strap.

A harness that has arrested a fall is immediately withdrawn from use and destroyed!

Risk matrix for erection work at height

RISK / SOURCE	CONSEQUENCE	CONTROL AND PROTECTION
Person falling from height	Severe multiple trauma, spinal fractures, death	100% tie-off, Y-lanyard, horizontal anchor line
Falling objects and tools	Injury to personnel at ground level	Helmets, protective canopies, tool lanyards, 15 cm toe boards
Scaffold/platform instability	Collapse of the structure with people and materials	Anchoring to the facade, diagonal bracing, formal acceptance
Weather (wind, thunderstorm, ice)	Blown off by a gust, loss of balance, slipping, lightning strike	Monitor forecasts, stop work at wind of 15 m/s or more (10 m/s for solid structures and panels)

The danger zone below work at height must be fenced off with warning tape and signs!

CRITICAL BAN: Violations during work at height



- Being at height without a correctly worn and adjusted full body harness and a locked carabiner.
- Wind speed of 15 m/s or more, or 10 m/s or more when erecting solid structures and panels.
- Onset of a thunderstorm, squall, heavy rain or snowfall, fog (visibility under 50 m) or icing of structures and ladders.
- No approved work execution plan for work at height (WEP) or rescue and evacuation plan (REP).
- Using homemade, uncertified or damaged platforms, barrels or stands instead of scaffolding.

An unprotected fall gives no second chance — safety at height is absolute!

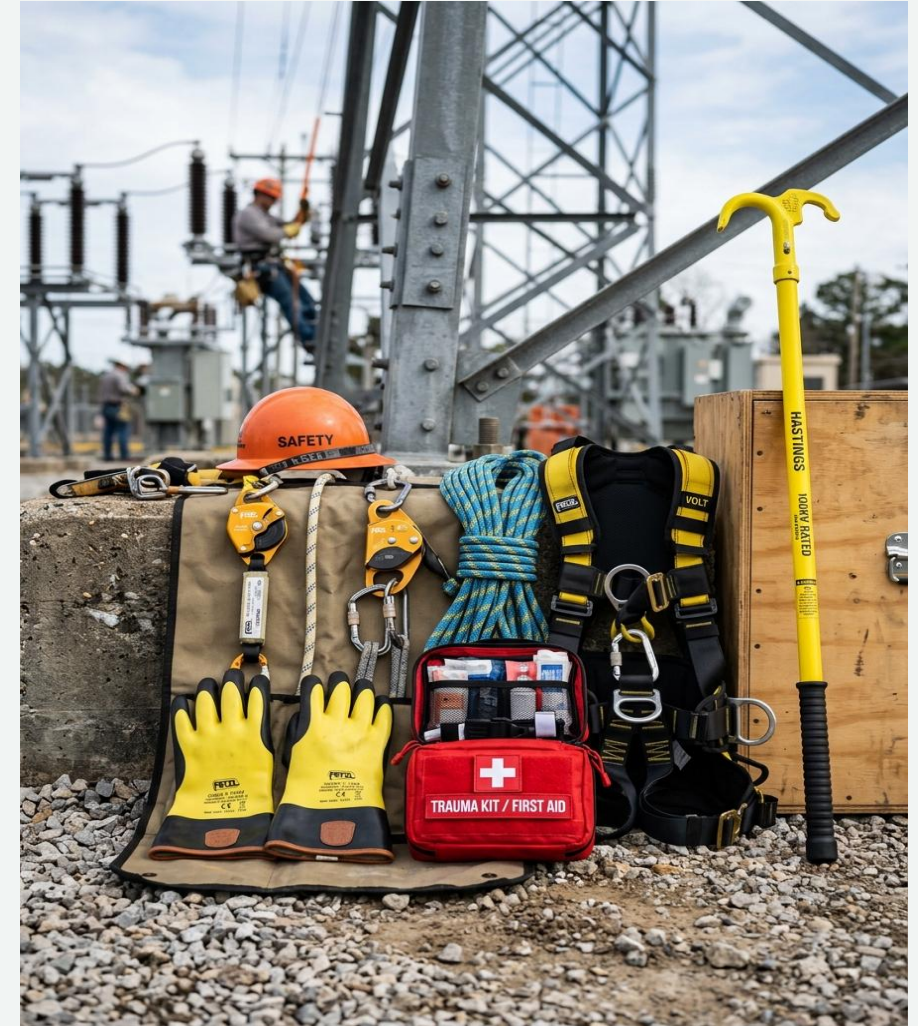


Combined work, rescue and first aid

Specifics of electrical work at height, preventing suspension trauma, emergency evacuation and first aid for victims.

Reference: REP • Cross-industry first aid instruction

Work at height • SafetyHUB





4 safety steps for work on OHL poles

1

Pole check

Check for wood rot or cracks in reinforced concrete, pole stability and guy wires.

2

100% tie-off

Use a flexible anchor line or a pole strap throughout the climb.

3

Safety zone

Keep safe clearances to adjacent lines that are live or carry induced voltage.

4

Earth the line

Fit portable earths on both sides of the worksite on all phase conductors.

Climbing angle poles from the inside of the conductor angle is strictly prohibited!



Aerial work platforms (AWP) near OHL

- **Work permit:** work within 30 metres of the outermost OHL conductor is done only under a work permit and a manager's supervision.
- **Chassis earthing:** the AWP vehicle body must be earthed with a flexible copper conductor to a fixed earth electrode or the substation earth.
- **100% tie-off:** an electrician in the basket must be clipped by the harness lanyard to the basket's dedicated certified anchor point.
- **No travel:** moving the AWP with people in the basket or with the boom raised is strictly prohibited!

Earthing the AWP body and clipping into the basket are mandatory before switching on the boom hydraulics!



4 stages of the rescue plan (REP)

1

Alarm and timer

Alert the supervisor at once and start timing.
Rescue time — up to 10 minutes!

2

Rig rescue kit

Attach the rescue device to a secure anchor above the victim or on an adjacent structure.

3

Hook and lower

Clip the rescue line to the suspended worker's harness, cut their lanyard, lower gently.

4

Shock control

Place them in the right position (sitting or semi-sitting!), call an ambulance.

The rescue and evacuation plan (REP) and rescue kit must be on site before work starts!



Suspension trauma in a harness

- **Injury mechanism:** thigh straps compress the femoral veins and blood pools in the legs, causing acute venous congestion.
- **Reduced blood return:** after 5–10 minutes of motionless hanging, brain hypoxia, loss of consciousness and cardiac arrest follow.
- **Self-help while suspended:** if conscious, actively move the legs (“pedal a bicycle”) and stand in the foot-loop stirrups.
- **GOLDEN RULE:** if conscious, do not lay the victim flat — keep them semi-sitting; if unconscious, lay them down and start CPR once breathing stops.

If conscious — semi-sitting, never flat on the back, and watched for at least 30 minutes.
If unconscious — lay down and start resuscitation!



Releasing a victim from current at height

- **Step 1. De-energise the installation:** switch off the nearest isolator, breaker or switch with the emergency stop.
- **Step 2. Prevent a fall:** remember that once the current is off, the victim will go limp and fall unless secured by fall protection!
- **Step 3. Insulate the rescuer:** if power cannot be cut quickly, use dielectric gloves, an insulating rod or an axe with a dry handle.
- **Step 4. Separate the victim:** pull the victim by dry clothing only, with one hand, without touching bare skin or metal.

When cutting the current to a victim at height, secure them from falling: the fall after a shock is deadlier than the current!



First aid: cardiopulmonary resuscitation (CPR)

- **Assessment:** check consciousness (call out, shake the shoulders) and breathing (the “listen — look — feel” rule for 10 seconds).
- **Chest compressions:** 30 presses on the centre of the breastbone, 5–6 cm deep, at 100–120 per minute with the heel of the hand.
- **Rescue breaths:** 2 mouth-to-mouth breaths through a barrier face shield after tilting the head back and pinching the nose.
- **Using an AED (defibrillator):** apply the automated defibrillator pads and follow the device's voice prompts.
- **Fractures and bleeding:** apply a pressure bandage to bleeding, splint fractures, do not reset bone fragments!

Start CPR immediately!
Every minute of delay cuts
the brain's chance of
survival by 10%!
Ratio: 30 to 2!

CRITICAL STOP: 6 grounds for an absolute stop



- 1. Sudden appearance of voltage on disconnected live parts or sparking at the worksite.
- 2. Worsening weather: onset of a thunderstorm, wind of 15 m/s or more (10 m/s for solid structures), snowstorm or dense fog.
- 3. Detection of a defect or tear in the harness, lanyard, carabiner or anchor line during work.
- 4. Sudden illness of any crew member (dizziness, nausea, weakness, chest pain).
- 5. Forced departure of the work supervisor or the person providing continuous supervision.
- 6. An emergency, or tools or structures falling near people.

The right to stop work (“STOP CARD”) is every worker's sacred right, with no risk of punishment!



Final admission checklist: 6 safety conditions

- ✓ **Work permit issued** — Permit for work at height signed, rescue plan approved, task briefing given.
- ✓ **Weather permits work** — Wind below 15 m/s (below 10 m/s for solid structures), no thunderstorm, heavy precipitation or fog.
- ✓ **Harness and lanyards checked** — Webbing, stitching, metal hardware and fall indicator inspected before the shift, no defects.
- ✓ **Full body harness fitted** — 100% continuous tie-off, anchor point reliable (≥ 22 kN), twin-leg lanyard connected.
- ✓ **Tools secured** — Tools on tool lanyards, toe boards and protective canopies installed.
- ✓ **Rescue plan ready** — Evacuation kit on site, communications working, danger zone below fenced off.